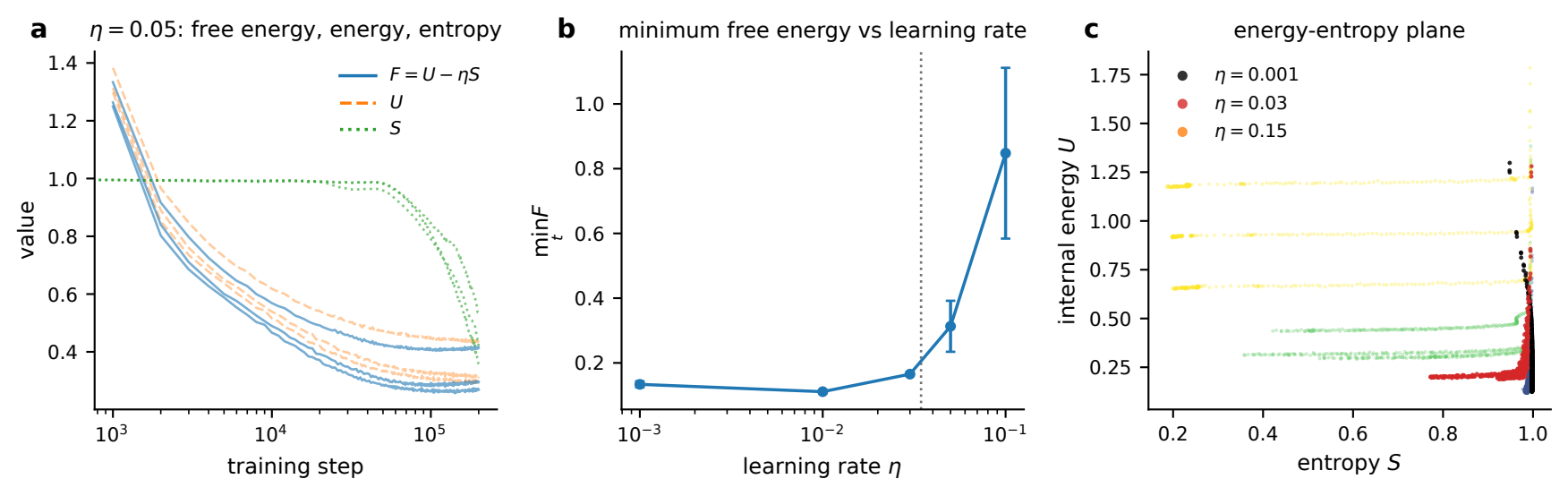

a
$\eta = 0.05$: free energy, energy, entropy
$F = U - \eta S$
U
S
value
training step
b
minimum free energy vs learning rate
$\min_t F$
learning rate η
c
energy-entropy plane
$\eta = 0.001$
$\eta = 0.03$
$\eta = 0.15$
internal energy U
entropy S